

Critical-scale growth bounds for congruence-preserving functions over finite fields

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Abstract

Let $R = \mathbb{F}_q[t]$. We prove that a function $f : R \rightarrow R$ preserving congruences modulo every irreducible polynomial is a polynomial map if $\deg f(A) \leq q^{\deg A}/(256q^2)$ for every input of sufficiently large degree. This replaces the factor q^n/n in the growth hypothesis of Bell and Nguyen by a positive constant times q^n . The proof uses integer-valued Carlitz digit polynomials, multiplied by a squarefree primorial, in the auxiliary-polynomial method. The rationality step is Bell and Nguyen's linear-growth theorem. For $\mathbb{F}_q$ -linear functions we prove the sharp threshold $q/(q-1)$ for $\limsup_n \deg f(t^n)/q^n$, and give a nonpolynomial example attaining it. We also give a Newton-coefficient formulation of the remaining sharp problem and an obstruction to a coefficient estimate using growth alone.

1 Results and their scope

Throughout, q is a prime power, $R = \mathbb{F}_q[t]$, and $K = \mathbb{F}_q(t)$. We set $\deg 0 = -\infty$. Write $V_n = \{A \in R : \deg A < n\}$, including zero, and let $\mathcal{P}$ be the set of monic irreducible polynomials in R . We say that $f : R \rightarrow R$ preserves prime congruences if

$$P \mid A - B \implies P \mid f(A) - f(B) \quad (A, B \in R, P \in \mathcal{P}). \quad (1)$$

A polynomial map is the evaluation of one polynomial in $K[X]$ on R . Its coefficients need not belong to R .

Bell and Nguyen [1, Theorem 1.4] proved polynomiality under $\deg f(A) < q^n/(27qn)$ for all sufficiently large $n = \deg A$. Their Section 4 asks whether the growth function q^n/n can be enlarged; Question 4.1 proposes every fixed constant below the primorial threshold. Our first theorem addresses the enlargement of the growth scale.

Theorem 1.1. *Let ℓ be the least positive integer such that $q^\ell \geq 16$, and set*

$$c_q = \frac{1}{16q^{\ell+1}}.$$

If $f : R \rightarrow R$ satisfies (1) and

$$\deg f(A) \leq c_q q^{\deg A} \quad (2)$$

for every input of sufficiently large degree, then f is a polynomial map. In particular, the bound $q^{\deg A}/(256q^2)$ suffices.

Corollary 1.2. *If f satisfies (1) and $\max_{\deg A=n} \deg f(A) = o(q^n)$, then f is a polynomial map.*

The constant in Theorem 1.1 is chosen for a transparent proof, rather than optimized. For $q = 2, 3, 4, 9$ the stated c_q is, respectively, $1/512, 1/1296, 1/1024, 1/11664$. Put

$$\Pi_n = \prod_{\substack{P \in \mathcal{P} \\ \deg P \leq n}} P, \quad d_n = \deg \Pi_n, \quad \Pi_0 = 1. \quad (3)$$

Then $d_n/q^n \rightarrow q/(q-1)$. The proposed sharp condition in [1, Question 4.1] is $\deg f(A) \leq (1-\varepsilon)d_n$ on every sufficiently large degree- n shell, for some $\varepsilon > 0$. Theorem 1.1 does not prove that full assertion.

Theorem 1.3. *Suppose $f : R \rightarrow R$ is $\mathbb{F}_q$ -linear and satisfies (1). If f is not a polynomial map, then*

$$\limsup_{n \rightarrow \infty} \frac{\deg f(t^n)}{q^n} \geq \frac{q}{q-1}.$$

This constant is sharp, even with the maximum over all inputs of degree n in the numerator. Thus Bell–Nguyen Question 4.1 holds for $\mathbb{F}_q$ -linear functions.

The interpolation polynomials below are classical. Wagner’s work [3, 4] describes related coefficient divisibility for functions preserving all polynomial moduli, including prime powers. Here only (1) is assumed, and the squarefree factor Π_n replaces the least common multiple with prime powers. Li and Sha [2] study congruence-preserving maps between finite residue rings. We use the classical interpolation tools to obtain a growth theorem for arbitrary maps on the infinite ring R .

2 Integral interpolation polynomials

Define

$$e_i(X) = \prod_{B \in V_i} (X - B), \quad H_i = e_i(t^i), \quad E_i(X) = \frac{e_i(X)}{H_i}. \quad (4)$$

In particular $e_0(X) = E_0(X) = X$ and $H_0 = 1$. These are the normalized linear Carlitz polynomials; see also [3, Section 2]. We give the needed elementary properties.

Lemma 2.1. *The polynomial E_i is $\mathbb{F}_q$ -linear, has X -degree q^i , and takes values in R on R . Moreover,*

$$\deg H_i = iq^i, \quad (5)$$

$$v_P(H_i) = \sum_{\substack{k \geq 1 \\ k \deg P \leq i}} q^{i-k \deg P}, \quad (6)$$

$$E_i(t^j) = 0 \ (j < i), \quad E_i(t^i) = 1, \quad (7)$$

$$\deg E_i(A) = (\deg A - i)q^i \quad (\deg A \geq i). \quad (8)$$

Proof. The decomposition $V_{i+1} = V_i + \mathbb{F}_q t^i$ gives inductively

$$e_{i+1}(X) = e_i(X)^q - H_i^{q-1} e_i(X).$$

This proves linearity and the asserted X -degree. The factors of H_i are exactly all monic degree- i polynomials, proving (5). For P of degree d , the number of those polynomials divisible by P^k is q^{i-kd} if $kd \leq i$ and zero otherwise. This proves (6).

For arbitrary A , if $e_i(A) = 0$ there is nothing to prove. Otherwise, for every k with $kd \leq i$, the map $V_i \rightarrow R/(P^k)$ is onto and has fibers of size q^{i-kd} . Therefore $v_P(e_i(A)) \geq v_P(H_i)$. This holds at every prime, so $E_i(A) \in R$. The triangular assertions follow from the definitions. If $\deg A \geq i$, every factor $A - B$, $B \in V_i$, has degree $\deg A$, proving (8). $\square$

Lemma 2.2. For $n \geq 1$,

$$q^n \leq d_n < \frac{q}{q-1} q^n, \quad d_n = \frac{q(q^n - 1)}{q-1} + O_q(q^{n/2}).$$

Proof. The squarefree polynomial $t^{q^n} - t$ divides Π_n , proving the lower bound. If I_j counts monic irreducibles of degree j , then $jI_j \leq q^j$, whence $d_n \leq \sum_{j=1}^n q^j < q^{n+1}/(q-1)$. From $jI_j = \sum_{a|j} \mu(a)q^{j/a}$, the error in replacing jI_j by q^j is at most $\sum_{r \leq j/2} q^r = O_q(q^{j/2})$. Summing gives the final assertion. $\square$

For $0 \leq j < q^s$, write $j = \sum_{i=0}^{s-1} j_i q^i$ with $0 \leq j_i < q$. Set

$$F_j(X) = \prod_{i=0}^{s-1} E_i(X)^{j_i}, \quad G_{s,j}(X) = \Pi_{s-1} F_j(X). \quad (9)$$

The first expression is independent of appended zero digits; $F_0 = 1$.

Lemma 2.3. The polynomials $G_{s,j}$, $0 \leq j < q^s$, are R -valued and preserve prime congruences. Their X -degrees are j , so they are linearly independent over K . Put $\kappa = (q+1)/(q-1)$. For $D \geq s-1$,

$$\deg G_{s,j}(A) < (D - s + \kappa)q^s \quad (A \in V_{D+1}). \quad (10)$$

Proof. Integrality follows from Lemma 2.1. If $\deg P \leq s-1$, every value of $G_{s,j}$ is divisible by P . If $\deg P > s-1$, all the H_i in (9) are P -units. Thus $G_{s,j} \in R_{(P)}[X]$ and preserves congruences modulo P . Also $\deg_X F_j = j$.

If a factor with $i > \deg A$ has positive exponent, the value is zero. In every other case, Lemmas 2.1 and 2.2 give, with $C = q/(q-1)$,

$$\begin{aligned} \deg G_{s,j}(A) &\leq d_{s-1} + \sum_{i=0}^{s-1} (q-1)(D-i)q^i \\ &< \frac{q^s}{q-1} + (D-s+C)q^s - D - C < (D-s+\kappa)q^s. \end{aligned}$$

Here the same strict bound for $d_0 = 0$ applies when $s = 1$. $\square$

3 The general growth theorem

We first recall the elementary vanishing principle underlying the Bell–Nguyen polynomial method [1, Lemma 2.1].

Lemma 3.1. Suppose $g : R \rightarrow R$ satisfies (1), vanishes on V_{M+1} , and $\deg g(A) < d_{\deg A}$ whenever $\deg A > M$. Then $g = 0$.

Proof. If not, choose A of least degree D with $g(A) \neq 0$. Then $D > M$. For each prime P of degree at most D , the remainder of A modulo P has degree less than D and hence has zero image. Condition (1) gives $\Pi_D \mid g(A)$, contradicting its degree bound. $\square$

Proposition 3.2. Under the hypotheses of Theorem 1.1, there is a nonzero $Q \in K[X, Y]$ such that $Q(A, f(A)) = 0$ for every $A \in R$.

Proof. Take M sufficiently large and put

$$s = M - \ell \geq 1, \quad T = \lfloor q^M/4 \rfloor, \quad J = 4q^{\ell+1}.$$

We may require $\deg f(A) \leq c_q q^M$ for every $A \in V_{M+1}$: the finitely many exceptions to (2) have bounded image degrees. Choose unknowns $u_{h,j,k} \in \mathbb{F}_q$ and form

$$Q(X, Y) = \sum_{h=0}^T \sum_{j=0}^{q^s-1} \sum_{k=0}^J u_{h,j,k} t^h G_{s,j}(X) Y^k. \quad (11)$$

Then $g(A) = Q(A, f(A))$ is R -valued and preserves prime congruences. Since $\ell \leq 4$, $\kappa \leq 3$, $q^\ell \geq 16$, and $Jc_q = 1/4$, Lemma 2.3 gives

$$\deg g(A) < \frac{q^M}{4} + (\ell + \kappa)q^{M-\ell} + \frac{q^M}{4} \leq \frac{15}{16}q^M < q^M \quad (A \in V_{M+1}). \quad (12)$$

Thus requiring $g(A) = 0$ for these q^{M+1} inputs imposes at most q^{2M+1} homogeneous linear equations over $\mathbb{F}_q$. The number of unknowns is strictly larger:

$$(T+1)q^{M-\ell}(J+1) > \frac{q^M}{4}q^{M-\ell}4q^{\ell+1} = q^{2M+1}.$$

A nonzero solution exists. It represents a nonzero polynomial Q : for each power of Y , the $G_{s,j}$ have distinct X -degrees, and $1, t, \dots, t^T$ are independent over $\mathbb{F}_q$.

For an input of degree $D = M + r$, $r \geq 0$, the same calculation gives

$$\frac{\deg g(A)}{q^D} < \frac{q^{-r}}{4} + (r + \ell + \kappa)q^{-\ell-r} + \frac{1}{4} \leq \frac{15}{16}. \quad (13)$$

The two r -dependent summands are nonincreasing: for $a \geq 1$, $(r + a + 1)/q \leq r + a$. By Lemma 2.2 we have $q^D \leq d_D$. Lemma 3.1 therefore gives $g = 0$. $\square$

Proof of Theorem 1.1. Clear denominators in the relation from Proposition 3.2, and write

$$Q = \sum_{k=0}^v P_k(X) Y^k, \quad P_k \in R[X], \quad P_v \neq 0.$$

Necessarily $v > 0$, since a nonzero polynomial in X cannot vanish on all of R . There are constants $a, b \geq 0$ such that

$$\deg P_k(A) \leq a \deg A + b$$

for every nonzero A and every k . If $P_v(A) \neq 0$ and $\deg f(A) > a \deg A + b$, the term $P_v(A)f(A)^v$ has strictly larger degree than every other term of $Q(A, f(A))$, a contradiction. Enlarge b to cover the finitely many roots of P_v and obtain $\deg f(t^n) = O(n)$ for $n \geq 0$.

Bell and Nguyen [1, Theorem 3.1] prove that a function satisfying (1) is a polynomial map if $\deg f(U^n) = O(n)$ for some $U \in R$ with $U' \neq 0$. Apply that theorem with $U = t$. Finally, minimality of ℓ gives $q^\ell < 16q$, and hence $c_q > 1/(256q^2)$. $\square$

The gain comes from (10): $q^{M-\ell}$ independent polynomials have value degrees $O(q^M)$ on V_{M+1} . Ordinary monomials of comparable X -degree have value degrees of order Mq^M . The rational denominators of the digit polynomials do not enter the value-height estimate, while the factor Π_{s-1} ensures the exact congruences needed for vanishing.

4 The sharp linear case

Every $\mathbb{F}_q$ -linear $f : R \rightarrow R$ has a unique locally finite expansion

$$f(A) = \sum_{i \geq 0} a_i E_i(A), \quad a_i \in R. \quad (14)$$

Indeed, recursively set $a_n = f(t^n) - \sum_{i < n} a_i E_i(t^n)$. Lemma 2.1 makes the coefficients integral. The resulting map agrees with f on the basis $1, t, t^2, \dots$, and $E_i(A) = 0$ for $i > \deg A$.

Proposition 4.1. *The function (14) preserves prime congruences if and only if $\Pi_i \mid a_i$ for every $i \geq 0$.*

Proof. Fix P of degree d . For $i < d$, H_i is a P -unit, so E_i is a polynomial over $R_{(P)}$. If $P \mid a_i$ for every $i \geq d$, all higher terms vanish modulo P , proving sufficiency.

Conversely, subtract $\sum_{i < d} a_i E_i$ from f . The result modulo P is a linear function on $R/(P)$ vanishing at the basis $1, t, \dots, t^{d-1}$, and hence vanishing everywhere. Evaluating successively at $t^d, t^{d+1}, \dots$ and using $E_i(t^i) = 1$ proves $P \mid a_i$ for every $i \geq d$. Apply this at all primes. $\square$

Lemma 4.2. *Write $\alpha_n = \deg a_n$, $\beta_n = \deg f(t^n)$, and $S_n = (q^n - 1)/(q - 1)$. Then*

$$\alpha_n \leq S_n + \max_{0 \leq j \leq n} (\beta_j - S_j). \quad (15)$$

Consequently,

$$\limsup_n \frac{\alpha_n}{q^n} \leq \max \left\{ \frac{1}{q-1}, \limsup_n \frac{\beta_n}{q^n} \right\}. \quad (16)$$

Proof. The recursion and (8) give $\alpha_n \leq \max\{\beta_n, \max_{i < n} (\alpha_i + (n-i)q^i)\}$. Since $(n-i)q^i \leq S_n - S_i$, induction proves (15). For the limiting assertion it suffices to consider a finite right side. Choose c strictly larger than it, and then choose $c_0 < c$ with $c_0 > 1/(q-1)$ such that eventually $\beta_j \leq c_0 q^j$. The finitely many earlier $\beta_j - S_j$ have an upper bound B . Equation (15) gives

$$\alpha_n \leq \max\{S_n + B, c_0 q^n\}$$

for all large n . Divide by q^n and let c decrease to the asserted bound. The identically zero map causes no exception with the convention $\deg 0 = -\infty$. $\square$

Proof of Theorem 1.3. If $\limsup \beta_n/q^n < q/(q-1)$, Lemmas 2.2 and 4.2 give $\deg a_n < d_n$ for all large n . Proposition 4.1 forces $a_n = 0$ there. Thus (14) is a finite polynomial.

For sharpness define

$$f_*(A) = \sum_{i \geq 0} \Pi_i E_i(A). \quad (17)$$

This map is well-defined, $\mathbb{F}_q$ -linear, and preserves prime congruences. Put $C = q/(q-1)$. For $\deg A = n$,

$$\deg f_*(A) \leq \max_{0 \leq i \leq n} \{d_i + (n-i)q^i\} \leq Cq^n,$$

because $d_i < Cq^i$ and $C + k \leq Cq^k$ for $k \geq 0$. For $i < n$ the same bound is at most $(C+1)q^{n-1}$: the sequence $(C+k)/q^k$ decreases for $k \geq 1$. At $A = t^n$ the term with $i = n$ has degree d_n , and $d_n/q^n \rightarrow C > (C+1)/q$. Eventually it strictly dominates all other terms. Therefore $\deg f_*(t^n) = d_n$ for all sufficiently large n . This gives the asserted sharp limit and proves nonpolynomiality, since a fixed polynomial map has value degrees $O(n)$ at t^n . $\square$

Wagner's characterization [3, Theorem 3.2] for the stronger all-moduli condition uses

$$L_i = \prod_{r=1}^i (t^{q^r} - t)$$

in place of Π_i . Proposition 4.1 is the squarefree counterpart. The boundary construction above and the degree estimate address the prime-congruence growth condition of [1].

5 Newton coefficients and the remaining threshold

Fix an ordering $\alpha_0 = 0, \alpha_1 = 1, \dots, \alpha_{q-1}$ of $\mathbb{F}_q$. For the base- q expansion $j = \sum_i j_i q^i$, put $A_j = \sum_i \alpha_{j_i} t^i$. The first q^m entries form V_m and $A_{q^m} = t^m$. Define

$$B_0(X) = 1, \quad B_n(X) = \frac{\prod_{j < n} (X - A_j)}{\prod_{j < n} (A_n - A_j)} \quad (n \geq 1). \quad (18)$$

Proposition 5.1. *Every function $f : R \rightarrow R$ has a unique locally finite expansion*

$$f(A) = \sum_{n \geq 0} b_n B_n(A), \quad b_n \in R.$$

It preserves prime congruences if and only if

$$\Pi_{\lfloor \log_q n \rfloor} \mid b_n \quad (n \geq 1). \quad (19)$$

There is no restriction on b_0 .

Proof. For P of degree d , the denominator in (18) has valuation

$$\sum_{k \geq 1} \left\lfloor \frac{n}{q^{kd}} \right\rfloor.$$

Indeed, each aligned block of q^{kd} consecutive entries runs through every residue modulo P^k . In the final incomplete block preceding A_n , none has its residue: their differences from A_n are nonzero and have degree less than kd . The numerator at any input has at least the same valuation. Thus all B_n are R -valued. Since $B_n(A_n) = 1$ and $B_n(A_i) = 0$ for $i < n$, integral triangular recursion gives the expansion and uniqueness.

Fix P of degree d . Terms with $n < q^d$ have P -unit denominators. If $P \mid b_n$ for $n \geq q^d$, the other terms vanish pointwise modulo P , proving sufficiency. Conversely, subtract the initial sum with $n < q^d$. The resulting congruence-preserving function vanishes on the complete residue system $A_0, \dots, A_{q^d-1}$, hence everywhere modulo P . Successive evaluation at the remaining A_n gives $P \mid b_n$ for $n \geq q^d$. Taking all primes proves (19). $\square$

This statement concerns first powers of primes. It makes no assertion that the same ordering characterizes preservation of all prime powers; compare the local constructions in [2, Theorem 1.11].

The sharp general question is consequently a truncation problem: does (19), together with the strict subcritical bound $(1 - \varepsilon)d_n$ on degree- n inputs, force $b_n = 0$ eventually? A finite expansion is equivalent to a polynomial map. The next example shows why value growth by itself cannot supply the needed coefficient margin.

Proposition 5.2. *Let $\delta(A)$ be 1 at $A = 0$ and 0 elsewhere. Its coefficients in (18) satisfy, for $m \geq 1$,*

$$\deg b_{q^m} = \frac{q^{m+1} - q}{q - 1} - m, \quad \frac{\deg b_{q^m}}{d_m} \rightarrow 1.$$

Proof. The leading coefficient of interpolation at $A_0, \dots, A_n$ gives

$$b_n = \frac{\prod_{j < n} (A_n - A_j)}{\prod_{j=1}^n (-A_j)}.$$

For $n = q^m$, the numerator has degree mq^m . The denominator has degree $m + \sum_{r=0}^{m-1} r(q-1)q^r$. Subtraction gives the formula; Lemma 2.2 gives the limit. $\square$

The function δ fails (1), since $\delta(0) = 1$ and $\delta(P) = 0$. It is an obstruction to an estimate using growth alone, not a counterexample to the sharp question. A proof at every constant below $q/(q-1)$ must use the congruences and the general coefficient transform together, or supply a different rigidity argument.

6 Verification and further directions

The proofs hold for every prime power and do not depend on finite computations. The accompanying exact-arithmetic checker tests both prime and extension fields. The digit-basis audit covers 8,570 literal values over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_9$, including integrality, degrees, and congruences. It also checks 56,320 instances of the height and propagation inequalities and 2,816 dimension inequalities. A separate audit covers the additive criterion, its boundary example, and the general Newton criterion with positive and negative examples.

The main remaining problem is the sharp arbitrary-function threshold. The integer-valued auxiliary family suggests optimizing its value-height profile and replacing the rectangular set of indices in (11) by a weighted set. Those changes do not by themselves prove the sharp constant. Theorems 1.1 and 1.3, and their coefficient formulations, constitute one growth-rigidity result rather than separate solutions to the full question.

References

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